

Module 1 IITP-AIML-TA-II-2607

Module 1: Foundations of Data

Module	Module Name	Batch Name	Lecture Date	Session Day	Lecture Name	Week no.	Description	Instructor	No. of hours	Session Start Time	Session End Time	Session Type	Host Email	Attendee	Rating	TA
		IITP-AIML-TA-II-2607	30-07-2026	Thursday	Virtual Inaugural Session		Virtual Inaugural Session		1	20:00:00	21:00:00	Virtual Inaugural				
		IITP-AIML-TA-II-2607	31-07-2026	Friday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	01-08-2026	Saturday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	02-08-2026	Sunday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	03-08-2026	Monday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	04-08-2026	Tuesday	Academic Session 1 - The AI Landscape & Programming Foundations		Distinguish AI, ML, and GenAI using real-world examples; set up a working dev environment (VS Code, Colab, Git/GitHub); securely manage API keys in a project	Aswath Rao	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	05-08-2026	Wednesday	Academic session 2 - Python Fundamentals		Identify and declare variables using correct data types; construct expressions using arithmetic and comparison operators; write and execute programs using input/output and f-strings in Colab	Aswath Rao	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	06-08-2026	Thursday	Tutorial Session 1			PC	1	20:00:00	21:00:00	Tutorial session				
		IITP-AIML-TA-II-2607	07-08-2026	Friday	Program Coordinator Connect			PC	1	20:00:00	21:00:00	PC Connect				
		IITP-AIML-TA-II-2607	08-08-2026	Saturday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	09-08-2026	Sunday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	10-08-2026	Monday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	11-08-2026	Tuesday	Academic session 3 - Control Flow & Decision Making		Construct if/elif/else blocks to handle real decision scenarios; evaluate boolean expressions combining multiple conditions; trace execution through nested conditions and predict output	Aswath Rao	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	12-08-2026	Wednesday	Academic session 4 - Loops, Iteration & Repetitive Logic		Write for and while loops with correct termination conditions; apply break and continue to control loop flow; iterate over sequences and access elements by index	Aswath Rao	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	13-08-2026	Thursday	Tutorial Session 2			PC	1	20:00:00	21:00:00	Tutorial session				
		IITP-AIML-TA-II-2607	14-08-2026	Friday	Program Coordinator Connect			PC	1	20:00:00	21:00:00	PC Connect				
		IITP-AIML-TA-II-2607	15-08-2026	Saturday	HOLIDAY			No session				Holiday				
		IITP-AIML-TA-II-2607	16-08-2026	Sunday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	17-08-2026	Monday	Talking About Your Present (Live)			Carolina NS	1.5	20:00:00	21:30:00	CSBT				
		IITP-AIML-TA-II-2607	18-08-2026	Tuesday	Academic session 5 - Master class: Numbers, Logic & Structure — The Mathematical Language of Data		Understand why 0s and 1s can represent any decision a program makes Read and construct truth tables for compound logical expressions See Python's data structures (lists, dicts, sets) as implementations of mathematical sets and functions Understand what a "function" means mathematically before it means anything in code	Abhinandhan	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	19-08-2026	Wednesday	Academic session 6 - Writing Reusable Code with Functions		Define and call functions with parameters and return values; explain how scope affects variable access; refactor repeated code blocks into reusable modular functions	Aswath Rao	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	20-08-2026	Thursday	Tutorial Session 3			PC	1	20:00:00	21:00:00	Tutorial session				
		IITP-AIML-TA-II-2607	21-08-2026	Friday	Program Coordinator Connect			PC	1	20:00:00	21:00:00	PC Connect				
		IITP-AIML-TA-II-2607	22-08-2026	Saturday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	23-08-2026	Sunday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	24-08-2026	Monday	Academic session 7 - Python Data Structures		Create and perform operations on lists, dictionaries, tuples and sets; differentiate between mutable and immutable structures; select the appropriate data structure for a given problem	Aswath Rao	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	25-08-2026	Tuesday	Academic session 8 - File Handling, JSON & APIs		Read and write files safely using context managers; parse and generate JSON data; make API calls using requests; handle responses and apply ethical usage of API keys and rate limits	Aswath Rao	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	26-08-2026	Wednesday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	27-08-2026	Thursday	Tutorial Session 4			PC	1	20:00:00	21:00:00	Tutorial session				
		IITP-AIML-TA-II-2607	28-08-2026	Friday	Program Coordinator Connect			PC	1	20:00:00	21:00:00	PC Connect				
		IITP-AIML-TA-II-2607	29-08-2026	Saturday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	30-08-2026	Sunday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	31-08-2026	Monday	Talking About Past Experiences (Recorded)			Carolina NS	1.5	20:00:00	21:30:00	CSBT				
		IITP-AIML-TA-II-2607	01-09-2026	Tuesday	Academic session 9 - NumPy: Numerical Foundation		Create and manipulate NumPy arrays using indexing and slicing; perform element-wise arithmetic and broadcasting operations without explicit loops; reshape and flatten arrays for downstream use	Aswath Rao	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	02-09-2026	Wednesday	Academic session 10 - Pandas: Loading, Inspection & Filtering		Load a CSV into a DataFrame and inspect its structure using head(), info() and describe(); filter rows using boolean conditions and loc/iloc; identify data quality issues from inspection output	Aswath Rao	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	03-09-2026	Thursday	Tutorial Session 5			PC	1	20:00:00	21:00:00	Tutorial session				
		IITP-AIML-TA-II-2607	04-09-2026	Friday	Program Coordinator Connect			PC	1	20:00:00	21:00:00	PC Connect				
		IITP-AIML-TA-II-2607	05-09-2026	Saturday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	06-09-2026	Sunday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	07-09-2026	Monday	Talking About Your Future & Connecting Past, Present, And Future (Live)			Carolina NS	1.5	20:00:00	21:30:00	CSBT				
		IITP-AIML-TA-II-2607	08-09-2026	Tuesday	Academic session 11- Pandas: Aggregation, Groupby & Merging		Group and aggregate data to answer business questions using groupby and agg; handle missing values by dropping or filling appropriately; merge and concatenate multiple DataFrames correctly	Aswath Rao	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	09-09-2026	Wednesday	Academic session 12 - Master class: From Tables to Relationships - The Mathematics of Data Organisation		Understand why data visualisation is rooted in coordinate geometry Compute and interpret mean, variance, and standard deviation manually and explain what they reveal	Abhinandhan	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	10-09-2026	Thursday	Tutorial Session 6			PC	1	20:00:00	21:00:00	Tutorial session				
		IITP-AIML-TA-II-2607	11-09-2026	Friday	Program Coordinator Connect			PC	1	20:00:00	21:00:00	PC Connect				
		IITP-AIML-TA-II-2607	12-09-2026	Saturday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	13-09-2026	Sunday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	14-09-2026	Monday	Holiday - Ganesh Chaturthi			Holiday				Holiday				
		IITP-AIML-TA-II-2607	15-09-2026	Tuesday	Academic session 13 - Data Visualization		Create line, bar, scatter and histogram plots using Matplotlib; build basic interactive charts using Plotly; select the appropriate chart type based on the variable type and analytical goal	Aswath Rao	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	16-09-2026	Wednesday	Academic session 14 - EDA & Business Thinking		Execute a structured EDA on a raw dataset identifying distributions, outliers and correlations; interpret patterns and anomalies in data; connect data findings to business questions like funnel drop-off and retention	Aswath Rao	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	17-09-2026	Thursday	Tutorial Session 7			PC	1	20:00:00	21:00:00	Tutorial session				
		IITP-AIML-TA-II-2607	18-09-2026	Friday	Program Coordinator Connect			PC	1	20:00:00	21:00:00	PC Connect				
		IITP-AIML-TA-II-2607	19-09-2026	Saturday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	20-09-2026	Sunday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	21-09-2026	Monday	Applying Articles & Commonly Used Prepositions (Recorded)			Carolina NS	1.5	20:00:00	21:30:00	CSBT				
		IITP-AIML-TA-II-2607	22-09-2026	Tuesday	Academic session 15 - SQL with MySQL Workbench			Aswath Rao	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	23-09-2026	Wednesday	Academic session 16- Data Analysis with Spreadsheets			Aswath Rao	2	20:00:00	22:00:00	Academic				
		IITP-AIML-TA-II-2607	24-09-2026	Thursday	Revision Session - 1			PC	-	-	-	No session				
		IITP-AIML-TA-II-2607	25-09-2026	Friday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	26-09-2026	Saturday	Module 1 End Evaluation							Evaluation				
		IITP-AIML-TA-II-2607	27-09-2026	Sunday	No session		No session	No session				No session				
		IITP-AIML-TA-II-2607	28-09-2026	Monday	Forming Longer, Meaningful Sentences (Live)			Carolina NS	1.5	20:00:00	21:30:00	CSBT				

Module 2 ITP-AIML-TA-II-2607

Module 2: Classical ML

ITP-AIML-TA-II-2607	29-09-2026	Tuesday	Academic session 17 -ML Workflow & Problem Framing	Write SQL queries to filter, sort and group data in MySQL Workbench; join two or more tables using INNER, LEFT and RIGHT joins; choose the correct join type for a given multi-table scenario	Aswath Rao	2	20:00:00	22:00:00	Academic			
ITP-AIML-TA-II-2607	30-09-2026	Wednesday	Academic session 18 -Data Preparation for ML	Use VLOOKUP and XLOOKUP to retrieve values across sheets; build pivot tables to summarize and slice data; apply COUNTIF and conditional formatting to surface quick data insights	Aswath Rao	2	20:00:00	22:00:00	Academic			
ITP-AIML-TA-II-2607	01-10-2026	Thursday	Tutorial Session 8		Aswath Rao	1	20:00:00	21:00:00	Tutorial session			
ITP-AIML-TA-II-2607	02-10-2026	Friday	Program Coordinator Connect		PC	1	20:00:00	21:00:00	PC Connect			
ITP-AIML-TA-II-2607	03-10-2026	Saturday	No session		No session				No session			
ITP-AIML-TA-II-2607	04-10-2026	Sunday	No session		No session				No session			
ITP-AIML-TA-II-2607	05-10-2026	Monday	Describing Skills, People, and Situations using Adjectives (Recorded)		Carolina NS	1.5	20:00:00	21:30:00	CSBT			
ITP-AIML-TA-II-2607	06-10-2026	Tuesday	Academic session 19 -Master Class: The Mathematics Behind Learning: Lines, Curves & Errors	Distinguish supervised and unsupervised learning and frame a business problem as the correct ML problem type; split data using train_test_split and evaluate using cross_val_score; justify metric selection based on the problem type	Abhinandhan	2	20:00:00	22:00:00	Academic			
ITP-AIML-TA-II-2607	07-10-2026	Wednesday	Academic session 20 - Linear Regression	Build an end-to-end preprocessing pipeline using sklearn Pipeline and ColumnTransformer; apply correct encoding and scaling strategies for different feature types; identify and prevent data leakage within a preprocessing workflow	Aswath Rao	2	20:00:00	22:00:00	Academic			
ITP-AIML-TA-II-2607	08-10-2026	Thursday	Tutorial Session 9		Vidhi	1	20:00:00	21:00:00	Tutorial session			
ITP-AIML-TA-II-2607	09-10-2026	Friday	Program Coordinator Connect		No session				No session			
ITP-AIML-TA-II-2607	10-10-2026	Saturday	No session		No session				No session			
ITP-AIML-TA-II-2607	11-10-2026	Sunday	No session		No session				No session			
ITP-AIML-TA-II-2607	12-10-2026	Monday	Reducing Fillers and MTI (Live)		Carolina NS	1.5	20:00:00	21:30:00	CSBT			
ITP-AIML-TA-II-2607	13-10-2026	Tuesday	Academic session 21 - Regularization	1.Derive and interpret the equation of a line from two points 2.Understand what a residual is and why minimising the sum of squared residuals gives you the best fit line 3.Explain what a derivative represents geometrically without computation 4.Describe gradient descent using only the analogy of a hill and a blindfolded person 5.Connect the math of a line to how a machine "learns" from data	Aswath Rao	2	20:00:00	22:00:00	Academic			
ITP-AIML-TA-II-2607	14-10-2026	Wednesday	Academic session 22 - Logistic Regression	Train a LinearRegression model using scikit-learn and generate predictions; evaluate model performance using MAE, RMSE and R ² ; interpret coefficients and diagnose overfitting by comparing train vs test performance	Aswath Rao	2	20:00:00	22:00:00	Academic			
ITP-AIML-TA-II-2607	15-10-2026	Thursday	Tutorial Session 10		Vidhi	1	20:00:00	21:00:00	Tutorial session			
ITP-AIML-TA-II-2607	16-10-2026	Friday	Program Coordinator Connect		PC	1	20:00:00	21:00:00	PC Connect			
ITP-AIML-TA-II-2607	17-10-2026	Saturday	No session		No session				No session			
ITP-AIML-TA-II-2607	18-10-2026	Sunday	No session		No session				No session			
ITP-AIML-TA-II-2607	19-10-2026	Monday	Body Language and Presence While Speaking (Recorded)		Carolina NS	1.5	20:00:00	21:30:00	CSBT			
ITP-AIML-TA-II-2607	20-10-2026	Tuesday	Academic session 23 -Classification Metrics	Apply Ridge and Lasso regression to reduce overfitting and compare with unregularized baseline; tune the alpha hyperparameter and observe its effect on coefficients; explain the bias-variance tradeoff in plain language	Aswath Rao	2	20:00:00	22:00:00	Academic			
ITP-AIML-TA-II-2607	21-10-2026	Wednesday	Academic session 24 -Master Class Probability & Counting :The Mathematics of Uncertainty	Train a LogisticRegression model and interpret predicted probabilities using predict_proba(); adjust the classification threshold and explain its effect on predictions; distinguish binary and multiclass classification setups	Abhinandhan	2	20:00:00	22:00:00	Academic			
ITP-AIML-TA-II-2607	22-10-2026	Thursday	Tutorial Session 11		Vidhi	1	20:00:00	21:00:00	Tutorial session			
ITP-AIML-TA-II-2607	23-10-2026	Friday	Program Coordinator Connect		No session				No session			
ITP-AIML-TA-II-2607	24-10-2026	Saturday	No session		No session				No session			
ITP-AIML-TA-II-2607	25-10-2026	Sunday	No session		No session				No session			
ITP-AIML-TA-II-2607	26-10-2026	Monday	Expressing Opinions and Viewpoints - (Live)		Carolina NS	1.5	20:00:00	21:30:00	CSBT			
ITP-AIML-TA-II-2607	27-10-2026	Tuesday	Academic session 25 -Decision Trees	Compute and interpret all classification metrics using sklearn metrics; analyze the confusion matrix to explain false positives and false negatives; evaluate the precision-recall tradeoff and select the right metric for a given business scenario	Aswath Rao	2	20:00:00	22:00:00	Academic			
ITP-AIML-TA-II-2607	28-10-2026	Wednesday	Academic session 26 - Random Forests & Ensemble Methods	Define sample space, event, and probability as a ratio Compute P(A B) and P(A B) manually from a small table or Venn diagram Derive Bayes' Theorem from the definition of conditional probability — no shortcuts Work through a complete Bayes' example numerically (e.g., if 1% of population has disease, test is 90% accurate — what is P(disease positive test)?) Explain in plain language why conditional probability is different from joint probability	Aswath Rao	2	20:00:00	22:00:00	Academic			
ITP-AIML-TA-II-2607	29-10-2026	Thursday	Tutorial Session 12		Vidhi	1	20:00:00	21:00:00	Tutorial session			
ITP-AIML-TA-II-2607	30-10-2026	Friday	Program Coordinator Connect		PC	1	20:00:00	21:00:00	PC Connect			
ITP-AIML-TA-II-2607	31-10-2026	Saturday	No session		No session				No session			
ITP-AIML-TA-II-2607	01-11-2026	Sunday	No session		No session				No session			
ITP-AIML-TA-II-2607	02-11-2026	Monday	Asking Questions Effectively (Recorded)		Carolina NS	1.5	20:00:00	21:30:00	CSBT			
ITP-AIML-TA-II-2607	03-11-2026	Tuesday	Academic session 27- Model Validation & Leakage	Train a DecisionTreeClassifier and visualize it using plot_tree; analyze overfitting by comparing train vs test accuracy across depth values; explain a prediction path from root to leaf in plain language	Aswath Rao	2	20:00:00	22:00:00	Academic			
ITP-AIML-TA-II-2607	04-11-2026	Wednesday	Academic session 28 - Clustering, Model Selection & Explainability	Train a RandomForestClassifier and extract feature importances to explain model behavior; compare random forest vs single decision tree on performance and interpretability; save and load a trained model using joblib	Aswath Rao	2	20:00:00	22:00:00	Academic			
ITP-AIML-TA-II-2607	05-11-2026	Thursday	Revision Session - 2		Vidhi	1	20:00:00	21:00:00	Tutorial session			
ITP-AIML-TA-II-2607	06-11-2026	Friday	No session		No session				No session			
ITP-AIML-TA-II-2607	07-11-2026	Saturday	Module 2 End Evaluation		No session				Evaluation			
ITP-AIML-TA-II-2607	08-11-2026	Sunday	No session		No session				No session			
ITP-AIML-TA-II-2607	09-11-2026	Monday	Storytelling for Everyday and Professional Situations (Live)		Carolina NS	1.5	20:00:00	21:30:00	CSBT			
ITP-AIML-TA-II-2607	10-11-2026	Tuesday	Academic Session 29 - LLM Fundamentals: How Language Models Work	Implement k-fold and stratified k-fold cross-validation to produce reliable performance estimates; tune hyperparameters using GridSearchCV and interpret results; detect data leakage scenarios and design a leakage-free evaluation pipeline	Aswath Rao	2	20:00:00	22:00:00	Academic			
ITP-AIML-TA-II-2607	11-11-2026	Wednesday	Academic Session 30 - Calling LLMs with APIs	Run KMeans clustering and determine optimal k using the elbow method and silhouette score; compare multiple trained models using a structured metric table; explain model predictions and feature importance in plain business language	Aswath Rao	2	20:00:00	22:00:00	Academic			
ITP-AIML-TA-II-2607	12-11-2026	Thursday	Tutorial Session 13		Vidhi	1	20:00:00	21:00:00	Tutorial session			
ITP-AIML-TA-II-2607	13-11-2026	Friday	Program Coordinator Connect		PC	1	20:00:00	21:00:00	PC Connect			
ITP-AIML-TA-II-2607	14-11-2026	Saturday	No session		No session				No session			
ITP-AIML-TA-II-2607	15-11-2026	Sunday	No session		No session				No session			
ITP-AIML-TA-II-2607	16-11-2026	Monday	Core Workplace Skills (Recorded)		Carolina NS	1.5	20:00:00	21:30:00	CSBT			
ITP-AIML-TA-II-2607	17-11-2026	Tuesday	Academic Session 31 - Prompt Engineering & Reasoning	Design effective zero-shot and few-shot prompts for specific tasks using role prompting; apply chain-of-thought reasoning to improve multi-step problem solving; evaluate and iteratively refine prompts using a defined test set and failure analysis	Aswath Rao	2	20:00:00	22:00:00	Academic			

Module 3: GenAI & Agents

					Add, subtract, and scale vectors both geometrically (arrows) and numerically Compute the magnitude of a 2D and 3D vector using Pythagoras Calculate the dot product of two vectors by hand Use the dot product formula to find the angle between two vectors Explain geometrically what it means for two vectors to be perpendicular (dot product = 0)	Abhinandhan	2	20:00:00	22:00:00	Academic	
IITP-AIML-TA-II-2607	18-11-2026	Wednesday	Academic Session 32 - Master Class: Vectors & Linear Algebra	The Mathematics of Space and Direction							
IITP-AIML-TA-II-2607	19-11-2026	Thursday		Tutorial Session 14		Vishal	1	20:00:00	21:00:00	Tutorial session	
IITP-AIML-TA-II-2607	20-11-2026	Friday		No session		No session				No session	
IITP-AIML-TA-II-2607	21-11-2026	Saturday		No session		No session				No session	
IITP-AIML-TA-II-2607	22-11-2026	Sunday		No session		No session				No session	
IITP-AIML-TA-II-2607	23-11-2026	Monday	Having Professional Conversations (Live)				No session			No session	
IITP-AIML-TA-II-2607	24-11-2026	Tuesday	Academic Session 33- Structured Outputs with Pydantic			Define structured output schemas using Pydantic BaseModel; patch an OpenRouter client with instructor to enforce structured JSON responses; validate and parse structured LLM outputs and handle malformed response gracefully	Aswath Rao	2	20:00:00	22:00:00	Academic
IITP-AIML-TA-II-2607	25-11-2026	Wednesday	Academic session 34- Getting Started with LangChain			Set up LangChain with OpenRouter as the LLM backend and explain why frameworks like LangChain exist; build prompt templates using PromptTemplate and ChatPromptTemplate; chain multiple LLM calls using LLMChain and SequentialChain	Aswath Rao	2	20:00:00	22:00:00	Academic
IITP-AIML-TA-II-2607	26-11-2026	Thursday		Tutorial Session 15		Vishal	1	20:00:00	21:00:00	Tutorial session	
IITP-AIML-TA-II-2607	27-11-2026	Friday		Program Coordinator Connect		PC	1	20:00:00	21:00:00	PC Connect	
IITP-AIML-TA-II-2607	28-11-2026	Saturday		No session		No session				No session	
IITP-AIML-TA-II-2607	29-11-2026	Sunday		No session		No session				No session	
IITP-AIML-TA-II-2607	30-11-2026	Monday		No session		No session				No session	
IITP-AIML-TA-II-2607	01-12-2026	Tuesday	Academic Session 35 - Building AI Workflows in LangChain			Build extraction and summarization workflows using LangChain chains and output parsers; design a simple FAQ bot and CSV copilot using LangChain pipeline patterns; construct multi-step AI pipelines that pass structured outputs between steps	Aswath Rao	2	20:00:00	22:00:00	Academic
IITP-AIML-TA-II-2607	02-12-2026	Wednesday	Academic Session 36 -Master Class : Calculus Intuition - Change, Slope & Optimisation			Compute the average rate of change of a function between two points Explain what a derivative is using only the tangent line picture — no epsilon-delta Apply basic derivative rules to simple polynomials by hand (e.g., $f(x) = 3x^2 + 2x + 1$) Find the minimum or maximum of a simple function by setting its derivative to zero Explain in plain language why "setting the derivative to zero" finds the bottom of a curve	Abhinandhan	2	20:00:00	22:00:00	Academic
IITP-AIML-TA-II-2607	03-12-2026	Thursday		Tutorial Session 16		Vishal	1	20:00:00	21:00:00	Tutorial session	
IITP-AIML-TA-II-2607	04-12-2026	Friday		No session		No session				No session	
IITP-AIML-TA-II-2607	05-12-2026	Saturday		No session		No session				No session	
IITP-AIML-TA-II-2607	06-12-2026	Sunday		No session		No session				No session	
IITP-AIML-TA-II-2607	07-12-2026	Monday		No session		No session				No session	
IITP-AIML-TA-II-2607	08-12-2026	Tuesday	Academic Session 37- Text Embeddings & Semantic Search			Generate text embeddings locally using sentence-transformers without any paid API; compute cosine similarity between embeddings to find semantically related content; distinguish semantic search from keyword search and explain when each is appropriate	Aswath Rao	2	20:00:00	22:00:00	Academic
IITP-AIML-TA-II-2607	09-12-2026	Wednesday	Academic Session 38-Chunking, Vector Stores & Retrieval			Chunk documents using RecursiveCharacterTextSplitter with appropriate size and overlap; index document chunks into both ChromaDB and FAISS and compare their trade-offs; retrieve relevant chunks using similarity search and MMR retrieval	Aswath Rao	2	20:00:00	22:00:00	Academic
IITP-AIML-TA-II-2607	10-12-2026	Thursday		Tutorial Session 17		Vishal	1	20:00:00	21:00:00	Tutorial session	
IITP-AIML-TA-II-2607	11-12-2026	Friday		Program Coordinator Connect		PC	1	20:00:00	21:00:00	PC Connect	
IITP-AIML-TA-II-2607	12-12-2026	Saturday		No session		No session				No session	
IITP-AIML-TA-II-2607	13-12-2026	Sunday		No session		No session				No session	
IITP-AIML-TA-II-2607	14-12-2026	Monday		No session		No session				No session	
IITP-AIML-TA-II-2607	15-12-2026	Tuesday	Academic Session 39- Building a RAG Pipeline			Build an end-to-end RAG pipeline connecting a ChromaDB retriever to an LLM via LangChain; inject retrieved context into prompts to ground LLM responses in source documents; compare grounded vs ungrounded responses to analyze and reduce hallucination	Aswath Rao	2	20:00:00	22:00:00	Academic
IITP-AIML-TA-II-2607	16-12-2026	Wednesday	Academic Session 40- Building an AI Agent with LangChain			Define custom tools and connect them to a LangChain AgentExecutor using the ReAct pattern; build a multi-step agent that selects and calls tools based on input; construct a working mini-agent build — either a CSV Analyst or FAQ Helpdesk — and control its execution safely	Aswath Rao	2	20:00:00	22:00:00	Academic
IITP-AIML-TA-II-2607	17-12-2026	Thursday		Tutorial Session 18		Vishal	1	20:00:00	21:00:00	Tutorial session	
IITP-AIML-TA-II-2607	18-12-2026	Friday		No session		No session				No session	
IITP-AIML-TA-II-2607	19-12-2026	Saturday		No session		No session				No session	
IITP-AIML-TA-II-2607	20-12-2026	Sunday		No session		No session				No session	
IITP-AIML-TA-II-2607	21-12-2026	Monday		No session		No session				No session	
IITP-AIML-TA-II-2607	22-12-2026	Tuesday	Academic Session 41 - Agentic RAG, Evaluation & Guardrails			Wrap a ChromaDB retriever as a LangChain tool and integrate it into an AgentExecutor for agentic RAG; evaluate RAG pipeline quality using RAGAS metrics — answer correctness, context recall and faithfulness; implement guardrails and explain responsible AI principles around hallucination, privacy and reliability	Aswath Rao	2	20:00:00	22:00:00	Academic
IITP-AIML-TA-II-2607	23-12-2026	Wednesday	Academic Session 42 - Agent Orchestration with LangGraph			Design a multi-step agent workflow using LangGraph StateGraph with defined nodes and transitions; implement conditional edges to handle branching logic and retry handling across agent steps; compare LangGraph orchestration with LangChain AgentExecutor and justify when to use each	Aswath Rao	2	20:00:00	22:00:00	Academic
IITP-AIML-TA-II-2607	24-12-2026	Thursday		Revision Session - 3		Vishal	1	20:00:00	21:00:00	Tutorial session	
IITP-AIML-TA-II-2607	25-12-2026	Friday		No session		No session				No session	
IITP-AIML-TA-II-2607	26-12-2026	Saturday	Module 3 End Evaluation				No session			Evaluation	
IITP-AIML-TA-II-2607	27-12-2026	Monday									
IITP-AIML-TA-II-2607	28-12-2026	Tuesday									
IITP-AIML-TA-II-2607	29-12-2026	Wednesday									
IITP-AIML-TA-II-2607	30-12-2026	Thursday									
IITP-AIML-TA-II-2607	31-12-2026	Friday									
IITP-AIML-TA-II-2607	01-01-2027	Saturday									
IITP-AIML-TA-II-2607	02-01-2027	Sunday									
IITP-AIML-TA-II-2607	03-01-2027	Monday									